Junior Neighborhood Detectives

Science Clues & Logic Cases for Ages 7–9

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Twenty friendly cases. Observe. Check. Explain.

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This is a work of fiction. Names, characters, places, and events are imaginary. Any resemblance to actual persons or events is coincidental.

Activities are intended for educational entertainment. Use the supplied records; no physical experiments are required. Children should ask a trusted adult for help when needed.

Welcome to the club

A clue is something you can check. A guess is an idea you have not checked yet.

Good detectives use both. They make a guess, look at the clues, and change their minds when they need to.

Maya likes to look closely. Sam likes to try an idea. Jo asks, “Could another answer fit?” You can do all three.

Our club rule: solve the mix-up, not blame a person.

Your detective toolkit

Use a pencil and an eraser. Write, circle, or draw an answer. Ask a grown-up to read with you if you like.

Each case has four pages: read the story, investigate, check, and explain. Take a break whenever you want.

Hints begin on page 89. Explained answers begin on page 99. Help is not a penalty.

You can skip a case. The final chapter repeats every fact it needs.

Keep investigations safe

Use the printed records. Do not taste unknown liquids, touch unknown plants, handle small magnets, or test electricity or heat.

Never follow a clue into an unfamiliar place. Stay with a trusted grown-up outdoors.

Some records describe tests done by an adult. You are reading those results, not being asked to do the experiment.

A wrong guess is not a bad person. Check the evidence and try again.

Words that help

Observe: notice something carefully. Record: save what was observed.

Evidence: information that helps answer a question. Compare: look for differences or similarities.

Eliminate: rule out a choice using a clue. Explain: say why the evidence fits.

A fair comparison changes one thing while keeping the other listed conditions alike.

Try one together

Two envelopes are marked A and B. One holds a circle card; the other holds a square card. The note says A holds the square.

Question: Where is the circle?

Work it out: A is taken by the square. Each envelope holds just one card. B is the only place left.

Answer: the circle is in B. Notice that we did not guess from envelope size or color. We used the note and the one-card rule.

Your case map

Garden Mix-Ups: cases 1–4, pages 9–24.

Light on the Clues: cases 5–8, pages 25–40.

Test the Idea: cases 9–12, pages 41–56.

Routes and Records: cases 13–16, pages 57–72.

The Fair Display Mix-Up: cases 17–20, pages 73–88.

Hint desk: 89–98. Answer desk: 99–118. Reflection and feedback: 119–120.

Case 1: The Three Garden Pots

The neighborhood fair opens tomorrow. Maya, Sam, and Jo have three pots ready for the garden table. But their name cards are on the floor. “Mine had a circle sticker,” says Maya. “Mine did not have a triangle sticker,” says Sam. “Let’s write that down,” says Jo. Each child owns exactly one pot. Each pot belongs to exactly one child. All three remember correctly.

Pot A: circle sticker; height 8 centimeters.

Pot B: triangle sticker; height 12 centimeters.

Pot C: square sticker; height 10 centimeters.

Stickers and heights have not changed. Height is measured from table to rim. A centimeter measures length. Use these recorded numbers.

Investigate

Case 1 • Activity 1

Match Maya, Sam, and Jo to pots A, B, and C. Write each name and pot letter.

Use this space for your matches, notes, or drawing.

Check the clues

Case 1 • Activity 2

Put the pot letters in order from shortest to tallest. Whose pot is tallest? Does it have the triangle sticker?

Which number, record, or rule helped you?

Explain and close

Case 1 • Activity 3

Jo says Sam cannot own A even though it is not a triangle. Why? Choose: A already belongs to Maya / Sam dislikes circles / A is too short. Explain your evidence.

The cards fell down. Nobody needed to be blamed. Sam copies the list so the club has a backup.

Think carefully: Pot height is not plant height. A taller pot does not prove that its plant is older or healthier.

Case 2: The Morning Deliveries

Three helpers brought one item each. Jo wants to thank them in the right order. “I wrote the clues,” says Maya. Nobody has to guess.

The deliveries were a watering can, a seed bag, and a sign. Each arrived once.

The seed bag arrived after the watering can.

The sign arrived before the watering can.

Investigate

Case 2 • Activity 1

Write the items in delivery order: first, second, third.

Use this space for your matches, notes, or drawing.

Check the clues

Case 2 • Activity 2

Which arrived just before the seed bag?

Which number, record, or rule helped you?

Explain and close

Case 2 • Activity 3

Sam says the seed bag arrived first. Which clue rules that out?

Jo writes three thank-you notes. Every delivery helped.

Think carefully: An order tells us before and after. It does not tell us the number of minutes between deliveries.

Case 3: Labels in the Seed Tray

The club has three labeled envelopes. The plants have not grown yet. “We can use the labels,” says Sam. “We cannot identify a plant from a bare pot.”

Tray positions are left, middle, right. Each holds one envelope: bean, pea, or sunflower.

The bean is not in the middle or on the right.

The pea is immediately to the right of the sunflower.

Investigate

Case 3 • Activity 1

Place all three envelope names in the tray.

Use this space for your matches, notes, or drawing.

Check the clues

Case 3 • Activity 2

Which envelope is between the other two?

Which number, record, or rule helped you?

Explain and close

Case 3 • Activity 3

Can these clues tell us which seed will sprout first? Explain.

The envelopes stay closed. The club saves planting for a grown-up-led activity.

Think carefully: A label is a record. It is not proof that a seed has sprouted.

Case 4: The Tool Hooks

Three safe garden tools hang on numbered hooks. Their record cards fell off. Maya checks the club notebook.

Hooks 1, 2, and 3 each hold exactly one item: brush, scoop, or watering can.

The brush is on an odd-numbered hook. Odd hooks here are 1 and 3.

The watering can is on hook 1.

Investigate

Case 4 • Activity 1

Match each item to its hook.

Use this space for your matches, notes, or drawing.

Check the clues

Case 4 • Activity 2

Which item is on hook 2?

Which number, record, or rule helped you?

Explain and close

Case 4 • Activity 3

If the brush were on hook 1, which stated rule would be broken?

Jo clips the copied record above the hooks.

Think carefully: We can reason from a complete inventory. With an unlisted extra tool, we would need more information.

Case 5: The Shadow Screen

A grown-up made three shadow tests for the fair. The club reads the records. No lamp or experiment is needed.

Only one opaque card was used in each test. Opaque means light does not pass through it.

Test A: circle card. Test B: triangle card. Test C: square card.

In this setup, each card faced the lamp and made the matching outline on the screen.

The saved outline has exactly three straight sides.

Investigate

Case 5 • Activity 1

Which test made the saved outline?

Use this space for your matches, notes, or drawing.

Check the clues

Case 5 • Activity 2

How many straight sides distinguish the square from the triangle here?

Which number, record, or rule helped you?

Explain and close

Case 5 • Activity 3

Would this record tell us the real size of the card? Explain.

The club adds a clear label to the shadow display.

Think carefully: Shadow size depends on the setup. This simplified record establishes shape, not size.

Case 6: Mirror Messages

Sam invents a paper mirror code. “This is a rule in our notebook,” he says. “You do not need a real mirror.”

The code reverses the ORDER of letters, not the shape of each letter.

Example: POT becomes TOP.

The label reads: XOB.

The three storage choices are BOX, BAG, and TIN.

Investigate

Case 6 • Activity 1

Decode XOB. Which storage choice matches?

Use this space for your matches, notes, or drawing.

Check the clues

Case 6 • Activity 2

Encode BAG with the same rule.

Which number, record, or rule helped you?

Explain and close

Case 6 • Activity 3

Jo writes POT as OPT. Does that follow the rule? Explain.

Maya labels it “reverse-order code” so nobody mistakes it for a real mirror test.

Think carefully: This is a letter-order code. A real mirror also reverses letter shapes, which this puzzle does not simulate.

Case 7: Light Through the Window

Jo compares saved light-test cards. An adult used the same lamp and detector setting each time. The club reads the observations.

Card A: clear sheet; lamp clearly visible through it.

Card B: tracing sheet; light patch visible, but lamp outline unclear.

Card C: thick card; lamp light blocked in this test.

For this puzzle: transparent = clear view; translucent = blurred light; opaque = blocks the light.

Investigate

Case 7 • Activity 1

Match transparent, translucent, and opaque to A, B, and C.

Use this space for your matches, notes, or drawing.

Check the clues

Case 7 • Activity 2

Which card lets light through without a clear view?

Which number, record, or rule helped you?

Explain and close

Case 7 • Activity 3

Sam wants to claim every thin sheet is transparent. Do these three observations prove that?

The display gets three observation labels instead of a big guess.

Think carefully: Material, thickness, and conditions matter. Describe these samples rather than every possible sheet.

Case 8: A Shadow That Changed

Maya finds three photos of one display. The club does not need to recreate them.

Same card, screen, and lamp setting in all three tests. Only the card’s position between lamp and screen changed.

Position A, nearest lamp: recorded shadow height 18 cm.

Position B, middle: 12 cm.

Position C, nearest screen: 6 cm.

Investigate

Case 8 • Activity 1

Order A, B, and C from smallest recorded shadow to largest.

Use this space for your matches, notes, or drawing.

Check the clues

Case 8 • Activity 2

How much taller is A’s shadow than B’s?

Which number, record, or rule helped you?

Explain and close

Case 8 • Activity 3

Did the card itself grow between tests? What fact supports your answer?

Jo adds “same card” beneath the photos. That detail matters.

Think carefully: These results apply to the stated setup. They do not tell clock time or show an object growing.

Case 9: The Magnet Record

A grown-up tested three sealed display samples. The children only read the results. “Let’s not guess from the shiny surface,” says Jo.

Each sample was tested three times with the same magnet and setup.

Sample A: moved toward magnet in all three tests.

Sample B: did not move in all three tests.

Sample C: moved toward magnet in all three tests.

The mystery card belongs to a sample that did NOT move.

Investigate

Case 9 • Activity 1

Which sample matches the mystery card?

Use this space for your matches, notes, or drawing.

Check the clues

Case 9 • Activity 2

Which TWO samples behaved alike?

Which number, record, or rule helped you?

Explain and close

Case 9 • Activity 3

Does a shiny appearance prove a material will move toward this magnet? Explain using the record’s limits.

The club uses test results rather than sparkle to label the card.

Think carefully: Do not handle small magnets. These observations do not identify exact metals or prove that all metals are magnetic.

Case 10: The Mixed-Up Cup Cards

The science table has sealed observation cards, not cups to taste. Sam sorts the results from an adult’s test.

All tests used equal water amounts, the same temperature, and one minute of stirring.

Card A: added sugar; no grains visible afterward.

Card B: added sand; grains still visible at the bottom.

Card C: added salt; no grains visible afterward.

The lost label says: “Grains still visible after stirring.”

Investigate

Case 10 • Activity 1

Which card needs the lost label?

Use this space for your matches, notes, or drawing.

Check the clues

Case 10 • Activity 2

Which cards describe the same visible result?

Which number, record, or rule helped you?

Explain and close

Case 10 • Activity 3

Can “no grains visible” prove that nothing is in the water? Explain.

Jo adds the word “visible” to the display. It makes the observation more careful.

Think carefully: Never taste an unknown liquid. The book uses records only; invisible does not mean absent or safe.

Case 11: The Paper Bridge

Three paper bridges were tested by an adult. Maya studies the records before choosing a photo for the fair.

Same paper size, support gap, and identical small weights. Only fold pattern changed.

A: flat sheet; held 2 weights before bending.

B: zigzag fold; held 8.

C: curved fold; held 5.

For this display, choose the tested pattern that held the MOST weights.

Investigate

Case 11 • Activity 1

Which pattern should the club show?

Use this space for your matches, notes, or drawing.

Check the clues

Case 11 • Activity 2

How many more weights did B hold than A?

Which number, record, or rule helped you?

Explain and close

Case 11 • Activity 3

Sam says B will hold the most in every possible bridge test. Is that established?

The club shows the conditions beside the winning photo.

Think carefully: These are model records, not instructions for a bridge people can stand on.

Case 12: The Rolling Record

A grown-up released the same toy car from the same ramp mark onto three labeled surfaces. Jo reads the measured distances.

A: 30 cm. B: 18 cm. C: 24 cm.

Same car, ramp mark, measuring method, and no push. Only the surface changed.

The record does not name the surface materials.

Investigate

Case 12 • Activity 1

Which surface had the longest recorded roll?

Use this space for your matches, notes, or drawing.

Check the clues

Case 12 • Activity 2

Order the surfaces from shortest roll to longest.

Which number, record, or rule helped you?

Explain and close

Case 12 • Activity 3

Can you identify which surface was carpet from these numbers alone? Explain.

Sam writes “surface names needed” rather than inventing them.

Think carefully: A fair comparison keeps listed conditions alike. A single set of results does not establish every future roll.

Case 13: The Delivery Map

The club’s map is a small square. Maya wants the shortest route on the marked paths. Each path takes one step.

Four places: Gate (top left), Library (top right), Garden (bottom left), Hall (bottom right).

Paths: Gate–Library, Gate–Garden, Library–Hall, Garden–Hall. Paths work both ways. No diagonal paths.

The Gate–Library path is closed.

Start at Gate and finish at Hall.

Investigate

Case 13 • Activity 1

Write the shortest open route.

Use this space for your matches, notes, or drawing.

Check the clues

Case 13 • Activity 2

How many steps does that route take?

Which number, record, or rule helped you?

Explain and close

Case 13 • Activity 3

Does a route through Library work without first reaching Hall? Explain.

Jo draws the open route on the fair’s delivery card.

Think carefully: This is a fictional map, not a real travel route. Stay with a trusted adult outdoors.

Case 14: The Sticker Chart

Sam made a chart of puzzle votes. Each child voted exactly once. “Read the key first,” says Maya.

Key: each star stands for 2 votes. These are full stars only.

Garden puzzles: 3 stars.

Light puzzles: 2 stars.

Map puzzles: 4 stars.

Investigate

Case 14 • Activity 1

How many votes did each kind receive?

Use this space for your matches, notes, or drawing.

Check the clues

Case 14 • Activity 2

How many votes were cast altogether?

Which number, record, or rule helped you?

Explain and close

Case 14 • Activity 3

Which kind won, and by how many votes over the runner-up?

The club keeps every kind of puzzle. A vote is not a reason to exclude anyone.

Think carefully: This chart describes these voters only. It does not prove what every child prefers.

Case 15: The Quiet Setup Slot

The fair helpers need a quiet time together. Jo checks their schedules. All times below are in the morning.

Maya is free 9:00–10:00.

Sam is free 9:30–10:30.

Jo is free 9:15–9:45.

They need 15 whole minutes when all three are free.

Investigate

Case 15 • Activity 1

What is the shared interval when all three are free?

Use this space for your matches, notes, or drawing.

Check the clues

Case 15 • Activity 2

Does 9:15–9:30 work? Why or why not?

Which number, record, or rule helped you?

Explain and close

Case 15 • Activity 3

Does 9:30–9:45 give all 15 minutes?

Maya puts the agreed time on the club card.

Think carefully: This puzzle uses stated schedules. It does not ask you to infer travel time.

Case 16: The Symbol Key

A box label uses Jo’s invented picture code. Sam has the full key. No outside code knowledge is needed.

Circle = B. Triangle = A. Square = G. Star = T.

A sequence is read from left to right.

Label: circle, triangle, square.

Investigate

Case 16 • Activity 1

Decode the label.

Use this space for your matches, notes, or drawing.

Check the clues

Case 16 • Activity 2

Encode BAT by writing the three symbol names in order.

Which number, record, or rule helped you?

Explain and close

Case 16 • Activity 3

Does square, triangle, circle spell the same word as the label? Explain.

The club puts the key beside the label so every helper can read it.

Think carefully: This is an invented substitution key, not a secret language or a scientific measurement.

Case 17: The Fair Crate Route

Fair day arrives. A crate of display cards is at the Gate. A helper needs it at the Hall. The club checks the route card.

Places in a line: Gate — Garden — Hall — Library.

Only neighboring places are connected. Each path takes one step, both ways.

The Garden–Hall path is open. All other shown paths are also open.

The crate was moved exactly two steps, without turning back, starting at Gate.

Investigate

Case 17 • Activity 1

Where is the crate now?

Use this space for your matches, notes, or drawing.

Check the clues

Case 17 • Activity 2

Which place did it pass through?

Which number, record, or rule helped you?

Explain and close

Case 17 • Activity 3

Could it be at Library after only two steps? Show the needed step count.

The crate is safe at Hall. Next, the club will sort the display cards.

Think carefully: Recap for this linked chapter: the display crate is at Hall. Later cases state any facts they need again.

Case 18: The Three Display Cards

The crate at Hall holds three display cards. Their fronts look alike. The notebook lists test results, so nobody has to open the sealed samples.

Red-label card: sample moved toward the magnet.

Blue-label card: sample did not move toward the magnet.

Green-label card: sample moved toward the magnet.

The labels have their color NAMES printed in words.

The fair’s magnet display needs the one card whose sample did NOT move.

Investigate

Case 18 • Activity 1

Which label belongs on that display?

Use this space for your matches, notes, or drawing.

Check the clues

Case 18 • Activity 2

Which two labels have matching results?

Which number, record, or rule helped you?

Explain and close

Case 18 • Activity 3

Does this establish that the blue-label sample is plastic? Explain.

The blue-label card goes to the display. Its material stays unnamed.

Think carefully: No actual magnets are needed. Label color does not cause the recorded behavior.

Case 19: The Missing Name Strip

The Hall table has three finished exhibits. Each child made one. Their name strips slipped into an envelope. Jo reads the notes.

Exhibits: bridge, map, shadow. Each belongs to exactly one of Maya, Sam, and Jo.

Maya made the map.

Sam did not make the bridge.

All notes are correct.

Investigate

Case 19 • Activity 1

Match each child to an exhibit.

Use this space for your matches, notes, or drawing.

Check the clues

Case 19 • Activity 2

Which exhibit belongs to Jo?

Which number, record, or rule helped you?

Explain and close

Case 19 • Activity 3

Sam says Jo made the shadow. Which two clues together show that cannot work?

The names go back beside the exhibits. Everyone’s work gets credit.

Think carefully: These notes establish ownership for this case. Do not infer skill or preferences from who made an exhibit.

Case 20: The Fair Can Open

One final delivery card is missing its destination. The club puts together all the facts. Each fact below is enough to check without returning to earlier pages.

Stops in order: Gate — Garden — Hall — Library. Each neighboring path is one step, both ways.

The delivery started at Gate and moved exactly two steps without turning back.

The parcel holds the bridge exhibit. Jo made the bridge.

The recorded bridge test held 8 weights; the flat-sheet test held 2 under the same stated setup.

Investigate

Case 20 • Activity 1

Where is the parcel, and whose exhibit is inside?

Use this space for your matches, notes, or drawing.

Check the clues

Case 20 • Activity 2

How many more weights did the recorded bridge test hold than the flat-sheet test?

Which number, record, or rule helped you?

Explain and close

Case 20 • Activity 3

Write one conclusion you CAN support and one thing you CANNOT know from this record.

The fair opens. Maya, Sam, and Jo thank the helpers. “Best clue today?” asks Sam. “Checking before guessing,” says Jo.

Think carefully: A supported conclusion can be firm while other facts remain unknown. The open explanation task has more than one valid response.

Hint desk

Case 1: The Three Garden Pots

Hint 1: Start with Maya’s exact sticker.

Hint 2: After A is assigned, Sam must choose between B and C. Which is not a triangle?

Case 2: The Morning Deliveries

Hint 1: Find the item that has something before it and after it.

Hint 2: Put the can in the middle.

Hint desk

Case 3: Labels in the Seed Tray

Hint 1: Use both places that bean cannot occupy.

Hint 2: Only middle and right remain for sunflower and pea.

Case 4: The Tool Hooks

Hint 1: Assign the watering can first.

Hint 2: Which odd hook is still free?

Hint desk

Case 5: The Shadow Screen

Hint 1: Count sides rather than guessing from a name.

Hint 2: Compare three sides with the shapes listed.

Case 6: Mirror Messages

Hint 1: Start at the right-hand end.

Hint 2: X is the last letter after decoding.

Hint desk

Case 7: Light Through the Window

Hint 1: Use the supplied definitions.

Hint 2: A clear view differs from a patch of light.

Case 8: A Shadow That Changed

Hint 1: Use the shadow heights, not alphabetical order.

Hint 2: Compare the largest and middle numbers by subtraction.

Hint desk

Case 9: The Magnet Record

Hint 1: Look for NOT move.

Hint 2: Compare all three repeats for each sample.

Case 10: The Mixed-Up Cup Cards

Hint 1: Read the last line of each test.

Hint 2: Two records say no grains were visible; one says they remained.

Hint desk

Case 11: The Paper Bridge

Hint 1: Compare 2, 8, and 5.

Hint 2: The choice is about this test, not every bridge.

Case 12: The Rolling Record

Hint 1: Longest means the largest distance.

Hint 2: Do not add a material name that is missing from the evidence.

Hint desk

Case 13: The Delivery Map

Hint 1: Cross out the closed path.

Hint 2: Gate has just one open exit.

Case 14: The Sticker Chart

Hint 1: One star means two votes, not one.

Hint 2: Runner-up is the second-largest total.

Hint desk

Case 15: The Quiet Setup Slot

Hint 1: Start no earlier than the latest person’s start.

Hint 2: Finish no later than the earliest person’s finish.

Case 16: The Symbol Key

Hint 1: Replace one symbol at a time.

Hint 2: Do not reverse the sequence.

Hint desk

Case 17: The Fair Crate Route

Hint 1: Trace one path for each step.

Hint 2: Count edges, not the number of place names.

Case 18: The Three Display Cards

Hint 1: Use the words moved and did not move.

Hint 2: A behavior can rule out a match without naming the material.

Hint desk

Case 19: The Missing Name Strip

Hint 1: Assign Maya first.

Hint 2: Of bridge and shadow, which can Sam take?

Case 20: The Fair Can Open

Hint 1: Count from Gate: Garden is one step.

Hint 2: Use only the stated facts. No arrival time appears.

Answers • Case 1

The Three Garden Pots

Activity 1: Maya owns A, Sam owns C, and Jo owns B. Maya’s circle fixes A. Sam cannot take B or the already assigned A. Jo gets B.

Activity 2: A, C, B: 8 is less than 10, and 10 is less than 12. Jo’s pot is tallest. Yes, it has the triangle.

Activity 3: A already belongs to Maya. Shape preferences and required heights were never given.

Careful thinking: Pot height is not plant height. A taller pot does not prove that its plant is older or healthier.

Answers • Case 2

The Morning Deliveries

Activity 1: Sign, watering can, seed bag. The can must be between the other two items.

Activity 2: The watering can.

Activity 3: The seed bag arrived after the watering can, so it cannot be first.

Careful thinking: An order tells us before and after. It does not tell us the number of minutes between deliveries.

Answers • Case 3

Labels in the Seed Tray

Activity 1: Left: bean. Middle: sunflower. Right: pea. Bean must be left; the remaining adjacent order is sunflower then pea.

Activity 2: Sunflower.

Activity 3: No. The clues give positions, not germination times or growing conditions.

Careful thinking: A label is a record. It is not proof that a seed has sprouted.

Answers • Case 4

The Tool Hooks

Activity 1: Can on 1, scoop on 2, brush on 3. Hook 1 is taken, so brush uses the other odd hook.

Activity 2: Scoop.

Activity 3: Each hook holds one item; hook 1 already holds the can.

Careful thinking: We can reason from a complete inventory. With an unlisted extra tool, we would need more information.

Answers • Case 5

The Shadow Screen

Activity 1: B. A triangle has three straight sides.

Activity 2: A square has four; a triangle has three. The difference is one side.

Activity 3: No. The record gives shape, not card dimensions or lamp distances.

Careful thinking: Shadow size depends on the setup. This simplified record establishes shape, not size.

Answers • Case 6

Mirror Messages

Activity 1: BOX. Read XOB from its last letter to its first.

Activity 2: GAB.

Activity 3: No. Reversing all three letters gives TOP, not OPT.

Careful thinking: This is a letter-order code. A real mirror also reverses letter shapes, which this puzzle does not simulate.

Answers • Case 7

Light Through the Window

Activity 1: A transparent, B translucent, C opaque, using the supplied definitions.

Activity 2: B, the tracing sheet.

Activity 3: No. Thickness alone is not the tested rule, and only these three samples were observed.

Careful thinking: Material, thickness, and conditions matter. Describe these samples rather than every possible sheet.

Answers • Case 8

A Shadow That Changed

Activity 1: C, B, A: 6, 12, 18 cm.

Activity 2: 6 cm, because 18 minus 12 is 6.

Activity 3: No. The record explicitly says the same card was used and only its position changed.

Careful thinking: These results apply to the stated setup. They do not tell clock time or show an object growing.

Answers • Case 9

The Magnet Record

Activity 1: B is the only sample recorded as not moving.

Activity 2: A and C both moved in all three tests.

Activity 3: No. No appearance comparison was recorded, and not every shiny material behaves alike.

Careful thinking: Do not handle small magnets. These observations do not identify exact metals or prove that all metals are magnetic.

Answers • Case 10

The Mixed-Up Cup Cards

Activity 1: B, the sand observation.

Activity 2: A and C.

Activity 3: No. Something can dissolve and still be present even when separate grains are not visible.

Careful thinking: Never taste an unknown liquid. The book uses records only; invisible does not mean absent or safe.

Answers • Case 11

The Paper Bridge

Activity 1: B, the zigzag fold, held 8, the largest count.

Activity 2: 6 more: 8 minus 2.

Activity 3: No. These tests cover only the supplied materials, gap, and weights. Different conditions need testing.

Careful thinking: These are model records, not instructions for a bridge people can stand on.

Answers • Case 12

The Rolling Record

Activity 1: A, with 30 cm.

Activity 2: B, C, A: 18, 24, 30 cm.

Activity 3: No. The materials are not named; a distance does not uniquely identify a material.

Careful thinking: A fair comparison keeps listed conditions alike. A single set of results does not establish every future roll.

Answers • Case 13

The Delivery Map

Activity 1: Gate → Garden → Hall.

Activity 2: Two steps. There is no direct Gate–Hall path.

Activity 3: No. The direct Gate–Library path is closed. Reaching Library from Gate requires passing through Hall first.

Careful thinking: This is a fictional map, not a real travel route. Stay with a trusted adult outdoors.

Answers • Case 14

The Sticker Chart

Activity 1: Garden 6, Light 4, Map 8. Multiply each star count by 2.

Activity 2: 18: 6 + 4 + 8.

Activity 3: Map won with 8, two more than Garden’s 6.

Careful thinking: This chart describes these voters only. It does not prove what every child prefers.

Answers • Case 15

The Quiet Setup Slot

Activity 1: 9:30–9:45. The latest start is 9:30 and earliest finish is 9:45.

Activity 2: No. Sam is not free until 9:30.

Activity 3: Yes. All three are free throughout that 15-minute interval.

Careful thinking: This puzzle uses stated schedules. It does not ask you to infer travel time.

Answers • Case 16

The Symbol Key

Activity 1: BAG. Replace each symbol with its key letter.

Activity 2: Circle, triangle, star.

Activity 3: No. It spells GAB. Order changes the result.

Careful thinking: This is an invented substitution key, not a secret language or a scientific measurement.

Answers • Case 17

The Fair Crate Route

Activity 1: Hall: Gate to Garden is one, Garden to Hall is two.

Activity 2: Garden.

Activity 3: No. Gate to Library needs three steps along this line.

Careful thinking: Recap for this linked chapter: the display crate is at Hall. Later cases state any facts they need again.

Answers • Case 18

The Three Display Cards

Activity 1: Blue-label card. It is the only non-moving result.

Activity 2: Red-label and Green-label.

Activity 3: No. Many materials may give that result. The record does not identify the sample.

Careful thinking: No actual magnets are needed. Label color does not cause the recorded behavior.

Answers • Case 19

The Missing Name Strip

Activity 1: Maya: map. Sam: shadow. Jo: bridge.

Activity 2: Bridge.

Activity 3: Maya already has map; Sam cannot have bridge, so Sam must have shadow. One-to-one ownership leaves bridge for Jo.

Careful thinking: These notes establish ownership for this case. Do not infer skill or preferences from who made an exhibit.

Answers • Case 20

The Fair Can Open

Activity 1: At Hall, with Jo’s bridge. Two steps along the line reach Hall; ownership is explicitly supplied.

Activity 2: 6 more: 8 minus 2.

Activity 3: Example supported: the parcel reached Hall. Example unknown: its arrival time. Other answers are valid if they respect the record. This last prompt intentionally allows several explanations.

Careful thinking: A supported conclusion can be firm while other facts remain unknown. The open explanation task has more than one valid response.

Your detective reflection

Which case made you change your mind? What clue helped?

Which did you enjoy most: matching, measurement, maps, codes, or testing an explanation?

Draw yourself checking a clue. You do not need a badge to think carefully.

A note for grown-ups

Read instructions together if helpful. Praise a clear explanation, not speed. Accept equivalent answers supported by the evidence.

The last case includes an open explanation task. The answer desk gives an example, not the only valid response.

Ask: What do you know? What are you guessing? What evidence would change your mind?

For development feedback, note the case number, confusing instruction, assistance needed, and any other answer that fits. Do not record identifying child information in a shared file.

Thank you for helping a young detective think with care.